Sahaj Bhadja

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Summary

Customer-focused applications engineer experienced in **system-level architecture, application tools, product validation, cross-platform testing, troubleshooting, and technical communication**, delivering practical software solutions across complex integration scenarios.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Database Systems, Web Technologies, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Software Engineering, Operating Systems, Object-Oriented Programming, Data Retrieval

Experience

USC Dornsife **LA, California**
Software Engineer *September 2025 – Present*
- Developed reliable application services processing 5,000+ monthly jobs with explicit state transitions, failure isolation, validation, and reconciliation, increasing concurrent throughput 3x while reliably sustaining 99%+ production reliability.
- Analyzed system requirements and component interactions for 500+ users, reducing dashboard latency from 1.2 seconds to 350 milliseconds through schema design, indexing, caching, measurement, and performance optimization.

Grad8 **LA, California**
Software Engineer Intern *June 2025 – August 2025*
- Translated instructor requirements into user-friendly React and Flask applications integrating multimodal AI, OCR, AWS RDS, and REST APIs, supporting 150+ users and processing 10,000+ daily requests at scale.
- Diagnosed ingestion, model, and API failures, validating boundary cases across 1,000+ files per batch with automated tests, diagnostics, fallback paths, and reliably stable performance under concurrent workloads.

USC AD Labs **Los Angeles, CA**
ML Research Assistant *January 2025 – May 2025*
- Developed Python and PyTorch validation pipelines processing 100,000+ trajectory records, improving long-horizon planning 18% through system modeling, simulation, debugging, reproducible testing, and iterative performance analysis at scale.
- Evaluated closed-loop interactions across ego, road, and nearby-agent signals, improving stability 15% while clearly documenting technical tradeoffs, boundary scenarios, experimental findings, and integration recommendations for senior researchers.

Aasma Technology Solutions **India**
Software Developer Engineer - (SecureSync) *August 2023 – July 2024*
- Built cross-platform Java and Python application software, automating executable installation and 20+ unit and integration tests across Windows, macOS, and Linux to reliably validate consistent product behavior.
- Troubleshot customer-facing API integrations using filtering, pagination, caching, AWS Glue, and S3, improving response times 30%, reducing payload sizes 25%, and lowering technical debt 20% across releases.

LST Capitals **India**
Web Developer Intern *November 2022 – January 2023*
- Designed an end-to-end customer solution, mapped 12+ component integration points, and presented system architecture recommendations to the client before implementing cross-platform booking, payment, and notification workflows at scale.
- Developed and validated Flask APIs, Firebase synchronization, pricing, coupon, and payment interfaces across 50,000+ records, delivering sub-200-millisecond responses and reducing manual refund handling by 70% for customers.

Skills

Applications: System architecture, application development, integration, requirements analysis, technical support, documentation
Validation: Reproducible testing, boundary testing, unit testing, integration testing, test automation, debugging, troubleshooting
Languages: Python, Java, C, C++, JavaScript, TypeScript, SQL, Shell
Tools: REST APIs, OOP, algorithms, simulation, performance analysis, Git, CI/CD, PostgreSQL, AWS

Projects

CPU-scheduling Simulator Live | *Application Tool, Product Validation*
- Created a user-friendly tool implementing six C++ scheduling algorithms with automated measurements and Gantt visualization, validating boundary conditions across 1,000+ process combinations while reducing average turnaround time 25%.

Chat2SQL Code | *Customer Integration Tool*
- Developed Java and TypeScript software supporting 10+ databases, helping technical users explore schemas, generate queries, validate execution, and reliably troubleshoot Docker-based integrations across Windows, macOS, and Linux.

CaptionCraft - Image Caption Generator Live | *Comparative Product Analysis*
- Compared BLIP, GIT, ViT-GPT-2, and custom architectures across 10,000 images and 100,000 captions, validating performance specifications and improving measured accuracy from 81% to 96% through structured evaluation.